

Deep Learning-Based Parkinson's Disease Classification from T1-Weighted MRI Using Transfer Learning, Fusion Strategies, and Two-Cohort Participant-Level Evaluation

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Abstract—Parkinson's disease (PD) is a progressive neurodegenerative disorder for which reliable image-based classification remains challenging. This study evaluates transfer-learning and fusion strategies using T1-weighted magnetic resonance imaging (MRI) across two independent experimental cohorts.

The primary Parkinson's Progression Markers Initiative (PPMI) cohort contained 11,120 two-dimensional MRI slices from 52 participants, including 23 healthy controls and 29 participants with PD. MobileNetV2, VGG16, and DenseNet121 were evaluated individually and through feature-level early-fusion and decision-level late-fusion strategies. The best-performing model configurations identified on PPMI were subsequently retrained and independently evaluated on a combined NEUROCON–Tao Wu (NTW) cohort. The NTW cohort contained 3,320 MRI slices from 83 participants, including 36 healthy controls and 47 participants with PD, with 40 slices retained per participant.

On the matched PPMI image-level test set, VGG16 achieved an accuracy of 98.97%, an F1-score of 98.96%, and a ROC-AUC of 0.9994. Grid-optimized late fusion matched the 98.97% accuracy of VGG16 and achieved the highest image-level ROC-AUC of 0.9995. On eight completely unseen PPMI participants, DenseNet121, MobileNetV2, and the evaluated late-fusion methods achieved 87.50% accuracy and a ROC-AUC of 1.0000.

When the selected models were retrained on the NTW cohort, fine-tuned VGG16 achieved 87.95% slice-level accuracy, an F1-score of 90.38%, and a ROC-AUC of 0.9927. The VGG16–DenseNet121 concatenation model achieved the highest NTW slice-level accuracy of 93.98%, with an F1-score of 94.93% and a ROC-AUC of 0.9942. At the NTW participant level, the highest observed accuracy was 61.54% on 13 unseen participants. Fine-tuned VGG16 achieved 100.00% sensitivity but only 16.67% specificity, while the early-fusion models exhibited substantial sensitivity–specificity variation.

These findings demonstrate that high slice-level performance does not necessarily translate into reliable participant-level generalization. The variation between the PPMI and NTW experiments also demonstrates the importance of evaluating deep-learning models across independent cohorts. Representative Grad-CAM visualizations are included as a qualitative explainability analysis. Larger multicenter participant-disjoint studies remain necessary before clinical application.

Index Terms—Parkinson's disease, MRI, PPMI, NEUROCON, Tao Wu, NTW, deep learning, transfer learning, MobileNetV2, VGG16, DenseNet121, early fusion, late fusion, two-cohort evaluation, participant-level evaluation, Grad-CAM.

I. INTRODUCTION

Parkinson's disease (PD) is one of the most common progressive neurodegenerative disorders, affecting millions of people worldwide [1], [2], [3], [4]. It primarily results from the gradual degeneration of dopamine-producing neurons in the substantia nigra region of the brain, leading to both motor and non-motor symptoms. Common motor symptoms include resting tremor, muscle rigidity, bradykinesia, and postural instability, while non-motor symptoms may involve cognitive impairment, depression, sleep disturbances, and autonomic dysfunction. As the disease progresses, patients experience a significant decline in their quality of life, making early and accurate diagnosis essential for timely treatment and disease management.

Magnetic Resonance Imaging (MRI) has emerged as a valuable non-invasive imaging modality for investigating structural brain abnormalities associated with Parkinson's disease. However, identifying subtle changes in MRI scans through manual examination is challenging and often depends on clinical expertise. Recent advances in artificial intelligence, particularly deep learning, have demonstrated remarkable capabilities in automatically learning complex image representations directly from medical images without relying on handcrafted feature extraction techniques. Convolutional Neural Networks (CNNs) have therefore become increasingly popular for computer-aided diagnosis of neurological disorders.

Transfer learning has further accelerated the application of deep learning in medical imaging by utilizing knowledge learned from large-scale image datasets. Well-established architectures such as MobileNetV2, VGG16, and DenseNet121 have shown promising performance in various medical image classification tasks due to their strong feature extraction capabilities. Nevertheless, relying on a single deep learning model may limit the diversity of learned features. To overcome this limitation, fusion-based strategies have been introduced to combine complementary information extracted from multiple models. Early fusion integrates deep features before classification, whereas late fusion combines prediction probabilities from multiple independently trained models to improve classification robustness.

Although numerous studies have investigated deep learning techniques for Parkinson's disease classification, relatively few have systematically compared individual transfer learning

models with both early fusion and multiple late fusion strategies using the same MRI dataset. Furthermore, many existing approaches function as black-box models, providing limited insight into the image regions influencing model predictions. In clinical applications, model interpretability is an important requirement for increasing physician confidence and supporting reliable decision-making.

To address these challenges, this study presents a comprehensive deep-learning framework for Parkinson's disease classification using T1-weighted MRI images. The Parkinson's Progression Markers Initiative (PPMI) dataset was used as the primary experimental cohort. MobileNetV2, VGG16, and DenseNet121 were evaluated under a matched training protocol. Feature-level early-fusion and decision-level late-fusion strategies were subsequently investigated to determine whether combining complementary CNN representations or predictions improved Parkinson's disease classification.

After identifying the strongest model configurations in the PPMI experiments, the selected models were retrained and independently evaluated using a combined NEUROCON–Tao Wu (NTW) cohort. The NTW experiments used separate training and testing partitions and did not directly apply the PPMI-trained model weights to NTW images. Both slice-level and participant-level results were analyzed to evaluate whether the selected architectures produced consistent performance across two independent MRI cohorts. Finally, Gradient-weighted Class Activation Mapping (Grad-CAM) was used to provide representative qualitative visualizations of regions influencing VGG16 predictions.

The main contributions of this work are summarized as follows:

- Development of a Parkinson's disease classification framework based on MobileNetV2, VGG16, and DenseNet121.
- Investigation of feature-level early-fusion techniques based on feature concatenation and attention mechanisms.
- Evaluation of multiple decision-level late-fusion strategies, including equal averaging, weighted averaging, trainable-weight fusion, sample-wise attention, grid optimization, and logistic-regression stacking.
- Systematic comparison of individual, early-fusion, and late-fusion models using the PPMI MRI dataset.
- Participant-disjoint evaluation in which all MRI slices from each participant were assigned exclusively to the training, development, or testing subset.
- Independent retraining and evaluation of the selected best-performing model configurations using the combined NEUROCON–Tao Wu (NTW) cohort containing 3,320 MRI slices from 83 participants.
- Separate reporting of slice-level and participant-level NTW performance to assess whether the selected architectures produced consistent results across two independent MRI cohorts.
- Qualitative Grad-CAM analysis of representative VGG16 image-level predictions.

The remainder of this paper is organized as follows. Section II reviews related work on Parkinson's disease classification, transfer learning, fusion strategies, and explainable artificial

intelligence. Section III describes the PPMI and NTW datasets, preprocessing procedures, model architectures, and fusion strategies. Section IV presents the experimental setup and evaluation metrics. Section V presents and discusses the PPMI image-level, PPMI participant-level, NTW slice-level, and NTW participant-level results. Section VI concludes the paper, and Section VII presents future research directions.

II. RELATED WORK

A. Deep Learning for Parkinson's Disease Classification

Recent studies have applied machine learning and deep learning methods to MRI and Parkinson's Progression Markers Initiative data for diagnosis, disease staging, and progression analysis [5], [6], [7], [8], [9], [10].

MRI-based approaches are particularly valuable because structural brain images may contain subtle patterns associated with neurodegeneration. However, limited sample sizes, class imbalance, differences in imaging protocols, and the lack of external validation continue to affect the generalizability of many reported models. Recent studies have therefore explored multimodal deep learning, transfer learning, and explainable artificial intelligence to improve diagnostic performance and clinical interpretation [11], [12], [13], [14].

B. Transfer Learning for Medical Image Analysis

Transfer learning has become an effective approach for medical image classification, particularly when limited annotated data are available [15], [16], [17], [18]. Instead of training a deep neural network entirely from scratch, transfer learning adapts representations learned from large-scale datasets to a target medical-imaging task. This approach can reduce computational cost, accelerate convergence, and improve classification performance.

Several convolutional neural network (CNN) architectures have demonstrated strong performance in medical image analysis. VGG16 uses sequential convolutional layers to learn hierarchical image features [19]. DenseNet121 introduces dense connections between layers, enabling feature reuse and improved gradient propagation [20]. MobileNetV2 employs depthwise separable convolutions and inverted residual bottleneck blocks, providing a lightweight architecture with fewer parameters while maintaining competitive performance [21].

Several influential CNN architectures have shaped modern transfer-learning methods. AlexNet demonstrated the effectiveness of deep convolutional networks for large-scale image classification [22]. Residual connections subsequently enabled the development of substantially deeper networks [23]. MobileNet introduced computationally efficient depthwise separable convolutions [24], while EfficientNet proposed compound scaling across network depth, width, and input resolution [25].

Recent studies have successfully employed transfer learning for MRI-based Parkinson's disease diagnosis, reporting improved performance compared with conventional handcrafted feature extraction techniques. Fine-tuning pretrained CNN models enables the networks to capture disease-specific structural changes in brain MRI images while preserving the robust low-level visual representations learned from large natural

image datasets. Consequently, transfer learning has become an important component of modern computer-aided diagnosis systems for neurological disorders.

C. Early Fusion and Late Fusion Techniques

Fusion strategies combine complementary information produced by different models and are widely used in multimodal and ensemble learning [26], [27], [28]. Rather than relying on a single network, fusion methods can improve prediction stability by combining diverse feature representations or model outputs.

Early fusion combines feature representations extracted from multiple backbone networks before the final classification stage. The extracted feature vectors may be concatenated and subsequently refined using fully connected layers or attention mechanisms [29]. This enables the classifier to exploit complementary spatial and semantic information simultaneously.

Late fusion combines prediction probabilities generated by independently trained classifiers. Common approaches include equal averaging, weighted averaging, trainable global weighting, attention-based fusion, and stacking. Stacking uses the predictions of base models as inputs to a second-level classifier [30].

D. Explainable Artificial Intelligence

Although deep learning models have demonstrated strong classification performance, their decision-making process is often considered a black box, which can limit their acceptance in clinical practice. Explainable artificial intelligence techniques improve model transparency by showing how input features influence predictions [31], [32].

Gradient-weighted Class Activation Mapping (Grad-CAM) generates heatmaps that highlight image regions contributing strongly to a convolutional neural network prediction [33]. Other widely used explainability methods include Integrated Gradients, LIME, and SHAP [34], [35], [36]. In medical imaging, these visual explanations can help determine whether a model focuses on anatomically meaningful regions rather than irrelevant background information.

E. Research Gap

Despite significant advances in deep learning for Parkinson's disease diagnosis, several important research challenges remain. Most existing studies evaluate only a single transfer-learning architecture or a single fusion strategy, making it difficult to determine the relative strengths of different deep-learning frameworks under identical experimental conditions. Furthermore, many published studies rely exclusively on image-level evaluation, where MRI slices from the same participant may appear in different dataset partitions, potentially leading to optimistic performance estimates.

Another limitation of previous work is the limited investigation of decision-level fusion strategies. Most studies employ simple averaging or majority voting, whereas adaptive approaches such as trainable-weight fusion, attention-based fusion, grid-optimized fusion, and logistic-regression stacking

have rarely been evaluated systematically on Parkinson's disease MRI datasets. In addition, explainability is often limited or omitted, reducing confidence in the clinical applicability of the developed models.

To address these limitations, this study presents a unified deep-learning framework that systematically compares three pretrained CNN architectures—VGG16, DenseNet121, and MobileNetV2—together with feature-level early fusion and multiple decision-level late-fusion strategies. PPMI image-level and participant-disjoint evaluation protocols are investigated to assess the difference between slice-level discrimination and generalization to previously unseen participants.

The revised study additionally retrains and evaluates the selected architectures using the independent NEUROCON–Tao Wu (NTW) cohort. The NTW experiments provide a second-cohort assessment of architecture reproducibility under differences in participants, imaging sources, acquisition conditions, and data distributions. Because the selected architectures were retrained using NTW data, these experiments do not represent direct external validation of fixed PPMI-trained models.

III. MATERIALS AND METHODS

A. Dataset and Study Design

This study used two independent T1-weighted MRI cohorts. The Parkinson's Progression Markers Initiative (PPMI) dataset was used for the primary experiments. After the best-performing model configurations were identified using PPMI, the selected models were retrained and independently evaluated using a combined NEUROCON–Tao Wu (NTW) cohort.

1) *PPMI Cohort*: The primary experiments used T1-weighted MRI data obtained from the Parkinson's Progression Markers Initiative (PPMI) database [37], [38].

The original three-dimensional MRI scans were converted into two-dimensional image slices to enable training using convolutional neural networks. After preprocessing, the PPMI dataset contained 11,120 MRI slices, including 5,560 healthy-control images and 5,560 Parkinson's disease images. These images were obtained from 52 participants, comprising 23 healthy controls and 29 participants diagnosed with Parkinson's disease.

Each MRI slice inherited the diagnostic label of the participant from whom it was obtained. Therefore, every slice originating from a participant with Parkinson's disease was assigned the PD label, whereas slices originating from healthy participants were assigned the healthy-control label.

2) *NEUROCON–Tao Wu Cohort*: A second experimental cohort was created by combining T1-weighted MRI data from the NEUROCON and Tao Wu datasets [39], [40].

The original three-dimensional NTW MRI volumes were converted into two-dimensional PNG images using the same general volume-to-slice processing approach. Forty MRI slices were retained from every participant. The conversion produced 3,320 images without conversion errors.

The processed NTW cohort contained 1,440 healthy-control images from 36 participants and 1,880 Parkinson's disease images from 47 participants. Therefore, the complete NTW cohort consisted of 83 participants and 3,320 MRI slices.

The two source datasets were acquired using different MRI scanners and acquisition parameters. The NEUROCON scans were acquired using a 1.5-T Siemens Avanto scanner, whereas the Tao Wu scans were acquired using a Siemens Magnetom Trio 3-T scanner. Both datasets included T1-weighted MPRAGE scans, but their repetition times, echo times, and voxel dimensions differed [39], [40]. These acquisition differences introduce additional heterogeneity into the combined NTW cohort and may contribute to variation in model performance.

The selected best-performing model configurations from the PPMI experiments were retrained using NTW training data and independently evaluated using NTW test data. The PPMI-trained weights were not directly evaluated on NTW images. Consequently, the NTW experiment represents a second-cohort replication of the selected model architectures rather than direct external testing of the trained PPMI models.

Table I summarizes the participant and image distribution of the two experimental cohorts. The PPMI cohort was balanced at the image level, whereas the NTW cohort contained more Parkinson's disease participants and images than healthy controls. For this reason, accuracy was interpreted together with precision, recall, specificity, F1-score, and ROC-AUC.

B. PPMI Image-Level and Participant-Level Evaluation

Two experimental evaluation protocols were investigated in this study: an image-level split and a participant-level split.

1) *Image-Level Split*: In the image-level experiment, individual MRI slices were randomly divided into training, validation, and testing subsets. The exact MRI images included in the test subset were not used during model training. However, because a single MRI scan produces multiple slices, slices belonging to the same participant could occur in more than one subset.

This protocol evaluates the ability of the models to classify previously unseen MRI slices from the same dataset. However, it does not provide a completely independent evaluation of performance on previously unseen participants.

2) *Participant-Level Split*: To provide a stricter evaluation, a second experiment was conducted using a participant-disjoint split. The participant identifier was extracted from each image path using the PPMI_{PatientID} naming pattern.

All MRI slices containing the same participant identifier were grouped together before dataset splitting. Consequently, all slices belonging to a particular participant were assigned exclusively to either the training, development, or testing subset.

The final matched participant-level experiment assigned 36 participants to training, eight participants to development, and eight participants to testing. The training subset contained 16 healthy controls and 20 participants with Parkinson's disease. The development subset contained three healthy controls and five participants with Parkinson's disease, while the held-out testing subset contained four healthy controls and four participants with Parkinson's disease. The complete participant-level distribution is presented in Table II.

The participant identifiers assigned to the three subsets were compared before training. No participant identifier was

shared between the training, development, and testing subsets. Therefore, the participant-level testing set consisted entirely of participants whose MRI slices were not observed during model training.

C. NTW Image-Level and Participant-Level Evaluation

The selected best-performing model architectures from the PPMI experiments were retrained and evaluated using the NTW cohort. Two separate NTW evaluation protocols were used: an image-level protocol and a participant-level protocol.

1) *NTW Image-Level Evaluation*: For the NTW image-level experiment, individual MRI slices were divided into training, validation, and testing subsets. The held-out image-level test set contained 498 MRI slices. Based on the test-set confusion-matrix totals, the test set contained 216 healthy-control slices and 282 Parkinson's disease slices.

The test images were not used for model training. However, because the split was performed at the image level, different slices belonging to the same participant could occur in more than one partition. Therefore, the NTW image-level experiment measures the ability of the models to classify held-out MRI slices but does not provide a strictly participant-independent estimate of generalization.

2) *NTW Participant-Level Evaluation*: A separate participant-disjoint experiment was conducted to evaluate the selected models on completely unseen NTW participants. All 40 MRI slices belonging to the same participant were grouped before dataset splitting. Each participant and all associated slices were then assigned exclusively to the training, validation, or testing subset.

The held-out participant-level NTW test set contained 13 completely unseen participants and 520 MRI slices. It included six healthy controls and seven participants with Parkinson's disease. Because each participant contributed 40 slices, the 13 test participants produced a total of 520 test images.

No participant identifier was shared between the participant-level training, validation, and testing subsets. Slice probabilities belonging to the same participant were aggregated to obtain a single participant-level probability. A decision threshold of 0.5 was used for the VGG16 and VGG16-DenseNet121 configurations. For the MobileNetV2-VGG16 participant-level models, the selected thresholds were 0.4761 for concatenation and 0.8628 for attention. These thresholds were applied to the aggregated participant-level probabilities to obtain the final healthy-control or Parkinson's disease classifications.

The image-level and participant-level NTW experiments used separate data-partitioning protocols. Consequently, the 498-image test set used in the image-level experiment and the 520-image test set used in the participant-level experiment should not be interpreted as the same test set.

D. Image Preprocessing

All PPMI and NTW MRI slices were resized to 224×224 pixels and converted into three-channel images to satisfy the input requirements of the pretrained convolutional neural

TABLE I
SUMMARY OF THE PPMI AND NEUROCON-TAO WU MRI COHORTS

Cohort	HC Participants	PD Participants	Total Participants	HC Images	PD Images	Total Images
PPMI	23	29	52	5,560	5,560	11,120
NEUROCON-Tao Wu	36	47	83	1,440	1,880	3,320

TABLE II
MATCHED PARTICIPANT-LEVEL DATASET DISTRIBUTION

Subset	HC Participants	PD Participants	Total Participants	HC Images	PD Images	Total Images
Training	16	20	36	3,644	3,832	7,476
Development	3	5	8	768	960	1,728
Testing	4	4	8	1,148	768	1,916
Total	23	29	52	5,560	5,560	11,120

networks. Model-specific ImageNet preprocessing was applied separately for VGG16, DenseNet121, and MobileNetV2.

The two cohorts were processed and partitioned independently. Normalization parameters, data partitions, model checkpoints, and predictions generated for one cohort were not reused for the other cohort. This separation prevented information from the NTW test set from influencing model selection during the PPMI experiments.

Data augmentation was applied only to the training partition of each cohort. The augmentation operations included small random rotations, zoom transformations, and horizontal and vertical translations. These transformations were used to increase image variability and reduce overfitting while preserving the anatomical structure of the MRI scans. No augmentation was applied to validation or test images.

The participant-level models were trained using a batch size of 16. A fixed random seed of 42 was used during participant splitting and model initialization to improve the reproducibility of the experiments.

E. Transfer Learning Models

Three state-of-the-art convolutional neural network architectures were investigated in this study, namely MobileNetV2, VGG16, and DenseNet121. All models were initialized with ImageNet pretrained weights [41] and modified by replacing the original classification layer with a custom binary classifier for Parkinson's disease detection.

Global Average Pooling was employed to reduce the spatial feature maps before the fully connected classification layers. Batch normalization and dropout were incorporated to improve training stability and reduce overfitting [42], [43]. During the matched model-adaptation protocol, all three pretrained backbones were initially frozen while their custom classification heads were trained. Partial fine-tuning was subsequently performed by unfreezing the final eight VGG16 layers, the final 30 DenseNet121 layers, and the final 20 MobileNetV2 layers. This two-stage strategy preserved useful ImageNet representations while adapting higher-level features to the MRI classification task.

The PPMI and NTW experiments were conducted as independent training experiments. After the best-performing architectures were identified from the PPMI results, new instances

TABLE III
PARAMETER SUMMARY OF THE INDIVIDUAL TRANSFER-LEARNING MODELS

Model	Total Params	Trainable Params	Non-Trainable Params
VGG16	15,110,977	15,109,953	1,024
DenseNet121	7,334,465	7,250,049	84,416
MobileNetV2	2,620,481	361,729	2,258,752

of the selected architectures were initialized using ImageNet-pretrained weights and trained using the NTW training partition. The final PPMI-trained model weights were not used as the starting weights for the NTW experiments. Therefore, the NTW results evaluate the reproducibility of the selected architectures on a second cohort rather than direct transfer of a trained PPMI classifier.

Table III summarizes the total, trainable, and non-trainable parameter counts of the individual transfer-learning models. Complete layer-wise model summaries were generated using the TensorFlow/Keras `model.summary()` function. Optimizer-state parameters were excluded because they are not part of the model architecture.

The best-performing VGG16 configuration excluded the original ImageNet classification head and fine-tuned the pre-trained backbone. Global average pooling and a custom binary classification head were then added for Parkinson's disease classification.

F. Early Fusion Framework

The early-fusion experiments combined feature representations from two backbone networks at a time. The evaluated model pairs were VGG16 with DenseNet121 and MobileNetV2 with VGG16. For each pair, both direct feature concatenation and attention-based fusion were evaluated before the final binary classifier.

G. Late Fusion Strategies

Decision-level late fusion combined the prediction probabilities generated by VGG16, DenseNet121, and MobileNetV2. Let $p_m(x)$ represent the Parkinson's disease probability produced by model m for evaluation unit x .

For image-level evaluation, x represents one MRI slice. For participant-level evaluation, all slice-level probabilities belonging to participant j were aggregated using

$$\bar{p}_{m,j} = \frac{1}{n_j} \sum_{k=1}^{n_j} p_{m,jk}, \quad (1)$$

where n_j is the number of MRI slices belonging to participant j and $p_{m,jk}$ is the prediction of model m for slice k .

1) *Equal-Average Fusion*: Equal-average fusion assigned the same importance to all three base models:

$$p_{\text{avg}}(x) = \frac{1}{M} \sum_{m=1}^M p_m(x), \quad (2)$$

where x denotes the input MRI slice, $M = 3$ is the number of base models, m is the model index, and $p_m(x)$ is the predicted probability of Parkinson's disease produced by base model m . In this study, the three base models are VGG16, DenseNet121, and MobileNetV2. Therefore, $p_{\text{avg}}(x)$ represents the final probability obtained by assigning equal importance to the predictions of all three models.

2) *Manual Weighted Fusion*: Manual weighted fusion assigned fixed weights of 0.25, 0.50, and 0.25 to VGG16, DenseNet121, and MobileNetV2, respectively:

$$p_{\text{manual}}(x) = 0.25p_{\text{VGG}}(x) + 0.50p_{\text{Dense}}(x) + 0.25p_{\text{Mobile}}(x). \quad (3)$$

where $p_{\text{VGG}}(x)$, $p_{\text{Dense}}(x)$, and $p_{\text{Mobile}}(x)$ denote the Parkinson's disease probabilities predicted by VGG16, DenseNet121, and MobileNetV2, respectively. The coefficients 0.25, 0.50, and 0.25 are the manually assigned model weights. These weights sum to one, with DenseNet121 receiving the largest contribution. The resulting $p_{\text{manual}}(x)$ is the final manually weighted probability for input MRI slice x .

3) *Grid-Optimized Fusion*: Grid-optimized fusion selected the weight combination that maximized the development-set ROC-AUC:

$$\mathbf{w}^* = \arg \max_{\mathbf{w}} \text{AUC}_{\text{dev}} \left(\sum_{m=1}^M w_m p_m(x) \right), \quad (4)$$

$$\text{subject to } w_m \geq 0, \quad \sum_{m=1}^M w_m = 1. \quad (5)$$

Here, $\mathbf{w} = [w_1, w_2, \dots, w_M]$ denotes the vector of fusion weights, and $\mathbf{w}^*$ denotes the optimal weight vector selected using the development set. The operator $\arg \max_{\mathbf{w}}$ identifies the weight combination that produces the maximum development-set ROC-AUC, denoted by AUC_{dev} . The term $p_m(x)$ is the Parkinson's disease probability predicted by model m for input x , while w_m is the contribution assigned to that model. The constraints $w_m \geq 0$ and $\sum_{m=1}^M w_m = 1$ ensure that all model weights are nonnegative and normalized to sum to one.

4) *Global Trainable-Weight Fusion*: For global trainable-weight fusion, unconstrained model parameters a_m were converted into normalized weights using a softmax function:

$$w_m = \frac{\exp(a_m)}{\sum_{r=1}^M \exp(a_r)}. \quad (6)$$

The fused probability was calculated as

$$p_{\text{global}}(x) = \sum_{m=1}^M w_m p_m(x). \quad (7)$$

The learned weights remained constant across all evaluation samples.

5) *Sample-Wise Attention Fusion*: Sample-wise attention generated different fusion weights for each image or participant:

$$\mathbf{w}(x) = \text{softmax}(g_{\theta}(\mathbf{p}(x))), \quad (8)$$

where

$$\mathbf{p}(x) = [p_1(x), p_2(x), \dots, p_M(x)]^T. \quad (9)$$

The final attention-based probability was

$$p_{\text{att}}(x) = \sum_{m=1}^M w_m(x) p_m(x). \quad (10)$$

6) *Logistic-Regression Stacking*: For logistic-regression stacking, the standardized base-model probabilities were supplied to a meta-classifier:

$$p_{\text{stack}}(x) = \sigma \left(\beta_0 + \sum_{m=1}^M \beta_m z_m(x) \right), \quad (11)$$

where $z_m(x)$ represents the standardized probability generated by base model m , and

$$\sigma(u) = \frac{1}{1 + \exp(-u)} \quad (12)$$

is the sigmoid function. Grid optimization, global trainable weighting, sample-wise attention, and logistic-regression stacking were fitted using development data only.

H. Grad-CAM Explainability

Representative Grad-CAM visualizations from the fine-tuned VGG16 image-level model are presented as a qualitative explainability analysis. These visualizations are not interpreted as clinical or anatomical evidence without further expert validation.

IV. EXPERIMENTAL SETUP

A. Experimental Environment

All experiments were conducted using Python and the TensorFlow/Keras deep learning framework in the Google Colaboratory environment equipped with GPU acceleration. Image preprocessing, model training, evaluation, and visualization were implemented using TensorFlow, Keras, NumPy, Pandas, Scikit-learn, OpenCV, and Matplotlib libraries.

B. PPMI Matched Training Configuration

The final image-level and participant-level late-fusion experiments used the same model architectures and optimization schedule. The VGG16, DenseNet121, and MobileNetV2 backbones were initialized using ImageNet-pretrained weights with `include_top=False`. The original ImageNet classification layer was therefore removed and replaced with a custom binary classification head.

Training was conducted in two stages. During the first 10 epochs, the complete pretrained backbone was frozen and only the custom classification head was optimized. During the following 25 epochs, partial fine-tuning was performed by unfreezing the final layers of each backbone while keeping the earlier layers frozen. Specifically, the last eight VGG16 layers, the last 30 DenseNet121 layers, and the last 20 MobileNetV2 layers were made trainable. Therefore, each base CNN was trained for a total of 35 epochs.

The Adam optimizer was used with a learning rate of 1×10^{-4} during frozen-backbone training and 5×10^{-6} during fine-tuning. A batch size of 16 was used. Early stopping was disabled in the matched experiments so that all configured epochs were completed. ReduceLROnPlateau and ModelCheckpoint were retained.

The global trainable-weight fusion network and the sample-wise attention late-fusion network were each trained for 35 epochs. Equal-average fusion, manual weighted fusion, and grid-optimized fusion do not require neural-network training epochs. Logistic-regression stacking was fitted using development-set probabilities.

Binary cross-entropy was used as the loss function:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)], \quad (13)$$

where y_i represents the true class label and $\hat{y}_i$ represents the predicted probability of Parkinson's disease.

C. NTW Training Configuration

The NTW experiments were conducted after completion of the PPMI model comparison. Based on the PPMI findings, selected individual and early-fusion architectures were trained again using the NTW cohort. Each NTW model was initialized independently using ImageNet-pretrained backbone weights.

For the individual-model experiment, VGG16 was evaluated under both a frozen-backbone configuration and a fine-tuned-backbone configuration. In the frozen configuration, the pretrained convolutional backbone remained fixed while the custom binary classification head was trained. In the fine-tuned configuration, selected backbone layers were made trainable to adapt higher-level features to the NTW MRI cohort.

The NTW early-fusion experiments evaluated VGG16 with DenseNet121 and MobileNetV2 with VGG16. For each backbone pair, feature concatenation and attention-based fusion were evaluated. The concatenation method directly combined the pooled feature vectors, whereas the attention method learned data-dependent importance weights before classification.

The image-level and participant-level NTW experiments used their own training, validation, and testing partitions. Model checkpoints and fusion parameters were selected using NTW training and validation data only. The held-out NTW test data were used only for final performance evaluation.

D. Evaluation Metrics

Let TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively. Parkinson's disease was treated as the positive class.

Accuracy measures the proportion of all correctly classified observations:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \quad (14)$$

Precision measures the proportion of predicted Parkinson's disease cases that were correctly classified:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \quad (15)$$

Recall, also referred to as sensitivity, measures the proportion of actual Parkinson's disease cases correctly identified:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (16)$$

Specificity measures the proportion of healthy controls correctly identified:

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}. \quad (17)$$

The F1-score represents the harmonic mean of precision and recall:

$$F_1 = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (18)$$

For classification threshold t , the true-positive rate and false-positive rate were calculated as

$$\text{TPR}(t) = \frac{\text{TP}(t)}{\text{TP}(t) + \text{FN}(t)}, \quad \text{FPR}(t) = \frac{\text{FP}(t)}{\text{FP}(t) + \text{TN}(t)}. \quad (19)$$

The area under the receiver operating characteristic curve was defined as

$$\text{AUC} = \int_0^1 \text{TPR}(u) du, \quad (20)$$

where u denotes the false-positive rate. Confusion matrices and classification reports were also generated to examine model performance.

TABLE IV
MATCHED TRAINING CONFIGURATION FOR THE IMAGE-LEVEL AND PARTICIPANT-LEVEL LATE-FUSION EXPERIMENTS

Setting	Image-Level Experiment	Participant-Level Experiment
Evaluation unit	MRI slice	Participant
Train/development/test	60/20/20% of MRI slices	36/8/8 participants
Test set	2,224 MRI slices	8 participants
Participant overlap	Allowed across partitions	None
Input size	$224 \times 224 \times 3$	$224 \times 224 \times 3$
Batch size	16	16
Backbone configuration	include_top=False	include_top=False
Frozen-backbone training	10 epochs	10 epochs
Partial fine-tuning	25 epochs	25 epochs
Total base-model epochs	35	35
VGG16 layers fine-tuned	Last 8 layers	Last 8 layers
DenseNet121 layers fine-tuned	Last 30 layers	Last 30 layers
MobileNetV2 layers fine-tuned	Last 20 layers	Last 20 layers
Learning rates	10^{-4} and 5×10^{-6}	10^{-4} and 5×10^{-6}
Global trainable-weight fusion	35 epochs	35 epochs
Sample-wise attention fusion	35 epochs	35 epochs
Stacking	Logistic regression	Logistic regression

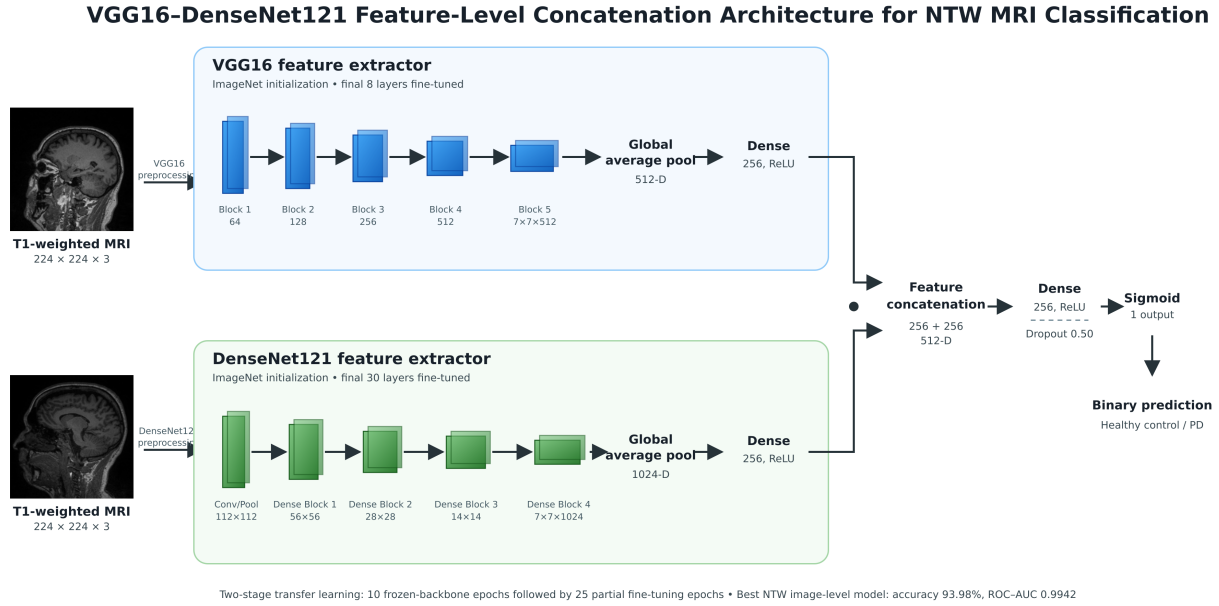


Fig. 1. Architecture of the VGG16-DenseNet121 feature-level concatenation model used for NTW image-level classification. The same T1-weighted MRI slice is supplied to both branches using model-specific preprocessing before feature extraction by the partially fine-tuned VGG16 and DenseNet121 backbones. Two representative MRI slices are displayed in the architecture diagram for visualization purposes. Global-average-pooled features are projected to 256 dimensions per branch and concatenated into a 512-dimensional feature vector. The fused representation is processed by a 256-unit dense layer, 0.50 dropout, and sigmoid binary classification.

V. RESULTS AND DISCUSSION

This section presents the results obtained from the two independent MRI cohorts. The PPMI experiments compare individual transfer-learning models, feature-level early fusion, and multiple decision-level late-fusion strategies. The NTW

experiments evaluate selected best-performing individual and early-fusion architectures after independent retraining on the NTW cohort.

Performance was assessed using accuracy, precision, recall, specificity, F1-score, and area under the receiver operating

TABLE V
MODEL CONFIGURATIONS EVALUATED ON THE NTW COHORT

Experiment	Architecture	Configuration
Individual model	VGG16	Frozen backbone
Individual model	VGG16	Fine-tuned backbone
Early fusion	VGG16 + DenseNet121	Concatenation
Early fusion	VGG16 + DenseNet121	Attention
Early fusion	MobileNetV2 + VGG16	Concatenation
Early fusion	MobileNetV2 + VGG16	Attention

TABLE VI
IMAGE-LEVEL CLASSIFICATION PERFORMANCE OF THE INDIVIDUAL
TRANSFER-LEARNING MODELS UNDER THE MATCHED PROTOCOL

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC
VGG16	98.97	99.19	98.74	98.96	0.9994
MobileNetV2	91.82	93.14	90.29	91.69	0.9780
DenseNet121	89.34	92.60	85.52	88.92	0.9702

characteristic curve (ROC–AUC). PPMI and NTW results are reported separately because the models were trained independently and the cohorts differed in participant composition and image distribution. Representative Grad-CAM visualizations from the fine-tuned PPMI VGG16 model are included as a qualitative explainability analysis.

A. PPMI Performance of Individual Transfer-Learning Models

Among the evaluated transfer-learning architectures, VGG16 achieved the strongest performance under the matched image-level evaluation protocol, obtaining an accuracy of 98.97%, a precision of 99.19%, a recall of 98.74%, an F1-score of 98.96%, and a ROC–AUC of 0.9994. The deep hierarchical feature representations learned by VGG16 effectively captured discriminative structural characteristics associated with Parkinson’s disease in T1-weighted MRI slices.

MobileNetV2 achieved competitive classification performance while maintaining the smallest computational complexity among the evaluated CNN models, making it attractive for deployment in resource-constrained environments. DenseNet121 also demonstrated strong performance; however, under the standardized matched training protocol, VGG16 consistently achieved superior overall classification accuracy and ROC–AUC.

These results demonstrate that transfer learning provides highly discriminative feature representations for Parkinson’s disease classification. Furthermore, the complementary prediction behavior of the three CNN architectures provides a strong motivation for the proposed early-fusion and late-fusion frameworks investigated in the following sections.

Figure 2, Figure 3, and Figure 4 present the training and validation accuracy and loss curves of the transfer-learning models under the image-level evaluation protocol.

The results in Table VI correspond to the image-level evaluation protocol. Although the test images were not used during training, MRI slices from the same participant could occur

in different subsets. Therefore, these results represent slice-level classification performance rather than generalization to completely unseen participants.

B. PPMI Performance of Early-Fusion Models

To investigate whether combining complementary feature representations could improve classification performance, several feature-level early-fusion frameworks were evaluated. Two model pairs—VGG16 with DenseNet121 and MobileNetV2 with VGG16—were tested using concatenation-based and attention-based fusion. The attention mechanism was used to emphasize informative features while suppressing redundant information.

Table VII compares the image-level performance of the concatenation-based and attention-based early-fusion models using a common test set of 1,668 MRI slices.

Attention-based feature fusion consistently outperformed simple feature concatenation for both evaluated backbone combinations, demonstrating that adaptive feature weighting enables the network to emphasize more informative representations while suppressing redundant features. Among the evaluated early-fusion architectures, the VGG16–DenseNet121 attention model achieved the highest classification performance with an accuracy of 97.66%, an F1-score of 97.65%, and a ROC–AUC of 0.9976.

Although early fusion successfully combined complementary feature representations from different CNN architectures, the best-performing early-fusion model did not surpass the standalone VGG16 model. Under the matched evaluation protocol, VGG16 achieved the highest image-level classification performance with an accuracy of 98.97% and a ROC–AUC of 0.9994, indicating that its hierarchical feature representations were already highly discriminative for Parkinson’s disease classification.

These findings suggest that feature-level fusion is beneficial when complementary representations exist between backbone networks; however, increased model complexity does not necessarily translate into higher classification accuracy. Nevertheless, the attention-based early-fusion framework demonstrated competitive performance while providing a flexible mechanism for combining complementary CNN features.

C. PPMI Matched Image-Level Late-Fusion Evaluation

The final matched image-level experiment was evaluated using 2,224 held-out MRI slices, comprising 1,112 healthy-control images and 1,112 Parkinson’s disease images. All methods were evaluated using the same test images and sample order.

Table VIII presents the performance of the three base CNNs trained using the matched 10+25 epoch schedule.

Table IX compares the six late-fusion strategies. Grid-optimized fusion achieved the highest ROC–AUC of 0.9995 and an accuracy of 98.97%. Global trainable-weight fusion produced the same threshold-based classification metrics. Sample-wise attention fusion and logistic-regression stacking each achieved an accuracy of 98.83%.

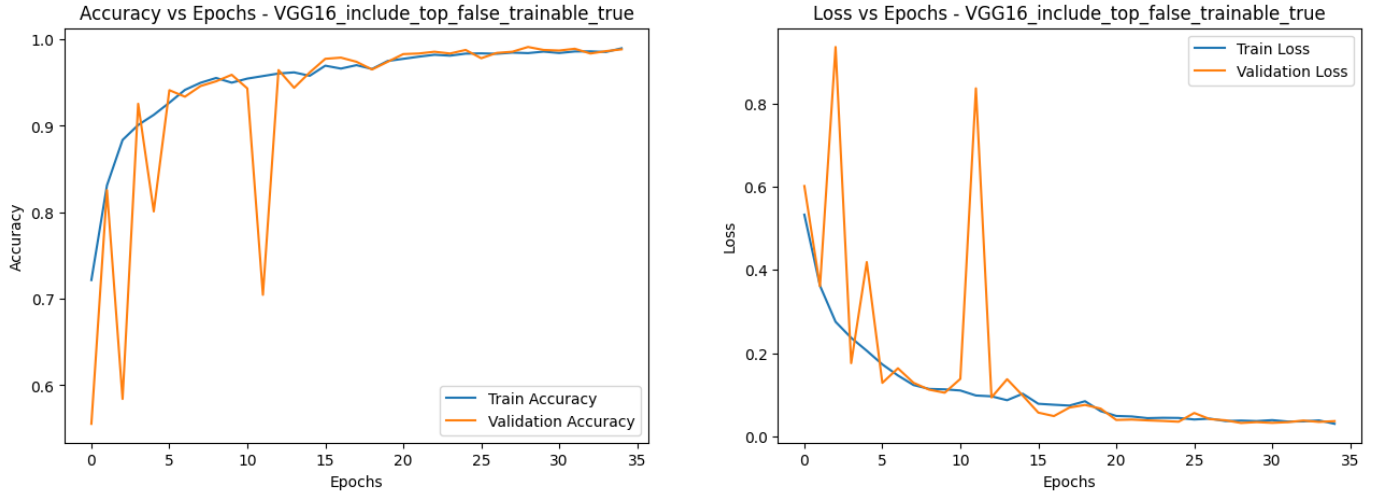


Fig. 2. Image-level VGG16 training performance: (a) training and validation accuracy and (b) training and validation loss.

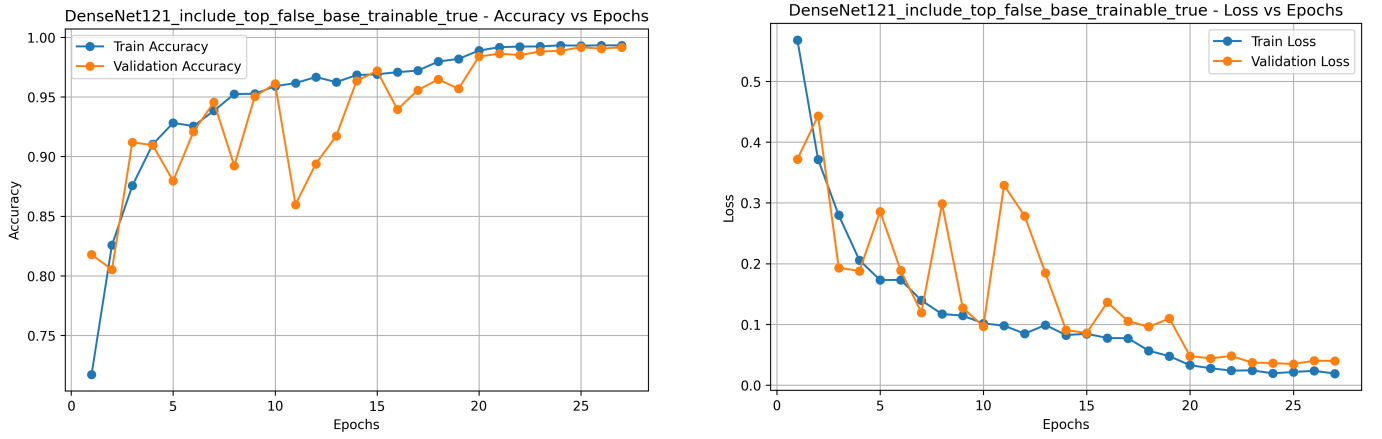


Fig. 3. Image-level DenseNet121 training performance: (a) training and validation accuracy and (b) training and validation loss.

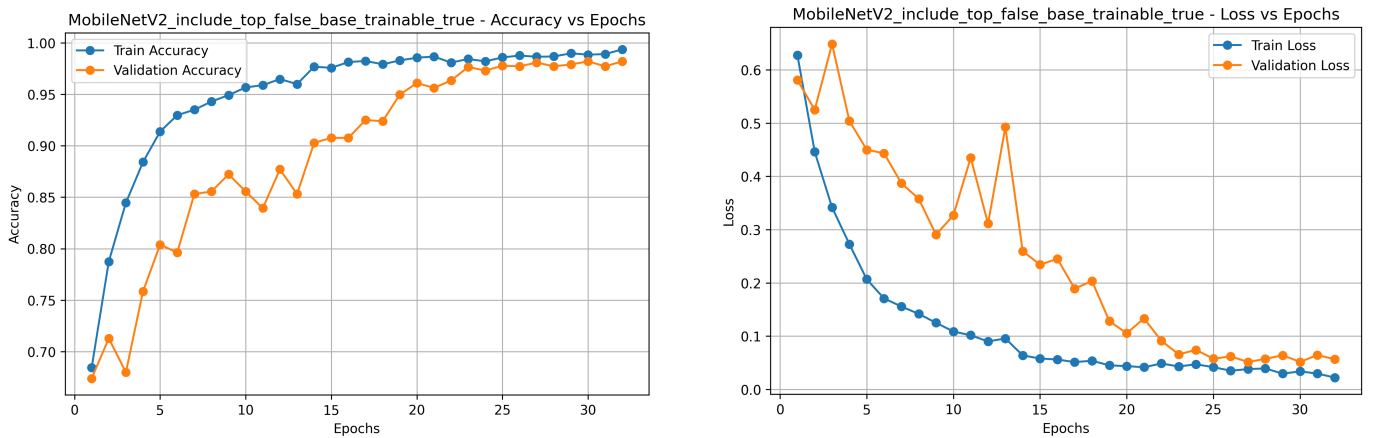


Fig. 4. Training and validation accuracy and loss curves for the fine-tuned MobileNetV2 model trained using the matched image-level protocol.

TABLE VII
IMAGE-LEVEL CLASSIFICATION PERFORMANCE OF THE EARLY-FUSION MODELS

Model Combination	Fusion Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC
VGG16 + DenseNet121	Concatenation	90.77	87.53	95.08	91.15	0.9702
VGG16 + DenseNet121	Attention	97.66	97.95	97.36	97.65	0.9976
MobileNetV2 + VGG16	Concatenation	81.83	82.66	80.58	81.60	0.8983
MobileNetV2 + VGG16	Attention	96.22	94.98	97.60	96.27	0.9931

TABLE VIII
MATCHED IMAGE-LEVEL PERFORMANCE OF THE INDIVIDUAL BASE CNN MODELS

Method	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1-score (%)	ROC-AUC
VGG16	98.97	99.19	98.74	99.19	98.96	0.9994
DenseNet121	89.34	92.60	85.52	93.17	88.92	0.9702
MobileNetV2	91.82	93.14	90.29	93.35	91.69	0.9780

TABLE IX
MATCHED IMAGE-LEVEL PERFORMANCE OF THE LATE-FUSION METHODS

Method	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1-score (%)	ROC-AUC
Equal-average fusion	97.48	97.91	97.03	97.93	97.47	0.9980
Manual weighted fusion	96.76	96.93	96.58	96.94	96.76	0.9959
Grid-optimized fusion	98.97	99.19	98.74	99.19	98.96	0.9995
Global trainable-weight fusion	98.97	99.19	98.74	99.19	98.96	0.9994
Sample-wise attention fusion	98.83	99.10	98.56	99.10	98.83	0.9995
Logistic-regression stacking	98.83	99.01	98.65	99.01	98.83	0.9994

The matched image-level evaluation demonstrates that decision-level fusion effectively combines the complementary strengths of the three transfer-learning models. Unlike early fusion, which merges intermediate feature representations, late fusion aggregates the final prediction probabilities generated by independently optimized CNNs. Consequently, each model contributes complementary diagnostic information while reducing the influence of individual prediction errors.

Among the evaluated approaches, the grid-optimized late-fusion strategy achieved the highest overall ROC-AUC while simultaneously maintaining excellent threshold-dependent classification performance. By optimizing the fusion weights on the development set, the proposed approach automatically emphasized the strongest individual classifiers while reducing the contribution of comparatively weaker predictions.

The global trainable-weight fusion and sample-wise attention fusion methods also achieved highly competitive performance. Unlike manually selected weights, these adaptive approaches learned the contribution of each CNN directly from the data, providing greater flexibility for heterogeneous MRI samples. Logistic-regression stacking likewise demonstrated strong performance by learning an optimal meta-classifier from the prediction probabilities generated by the individual transfer-learning models.

Overall, the late-fusion experiments indicate that combining complementary CNN predictions produces a more robust and stable decision-making framework than relying on a single classifier alone. These findings demonstrate the effectiveness of decision-level fusion for improving the reliability of Parkinson's disease classification from structural MRI images.

D. PPMI Matched Participant-Level Evaluation

The participant-level evaluation provides a substantially more rigorous assessment of model generalization than the image-level experiment because all MRI slices belonging to a participant were assigned exclusively to either the training, development, or testing subset. Consequently, the models were required to classify completely unseen participants rather than previously unseen MRI slices from participants encountered during training.

As expected, participant-level classification performance was lower than the corresponding image-level performance. This reduction does not indicate poorer model quality; rather, it reflects the increased difficulty of classifying completely unseen participants and the elimination of participant-level data leakage. Therefore, the participant-level results provide a more realistic estimate of the performance that can be expected in practical clinical applications.

Among the evaluated methods, the proposed late-fusion strategies maintained stable participant-level performance despite the substantially more challenging evaluation protocol. Aggregating prediction probabilities from multiple CNN architectures reduced the influence of individual model errors and improved decision robustness. These findings indicate that decision-level fusion is particularly beneficial when evaluating previously unseen participants, where greater variability exists between MRI scans.

Overall, the participant-level experiments demonstrate the importance of rigorous evaluation protocols in medical image analysis. While image-level experiments remain useful for benchmarking feature-learning capability, participant-level

evaluation provides a more reliable measure of clinical generalization and should therefore be preferred when assessing deep-learning models for Parkinson's disease diagnosis.

DenseNet121, MobileNetV2, and all six late-fusion methods achieved 87.50% accuracy, 100.00% precision, 75.00% recall, 100.00% specificity, an F1-score of 85.71%, and a ROC-AUC of 1.0000. VGG16 achieved 75.00% accuracy and a ROC-AUC of 0.9375.

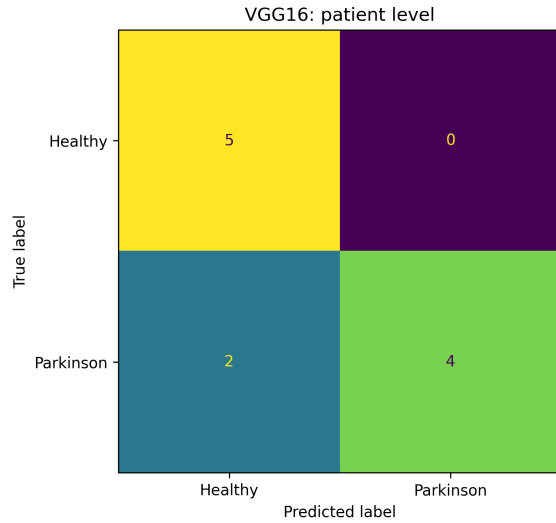


Fig. 6. Participant-level confusion matrix for the grid-optimized fusion method. Four healthy controls and three of four participants with Parkinson's disease were correctly classified.

TABLE X
MATCHED PARTICIPANT-LEVEL PERFORMANCE ON EIGHT COMPLETELY UNSEEN PARTICIPANTS

Method	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1-score (%)	ROC-AUC
VGG16	75.00	100.00	50.00	100.00	66.67	0.9375
DenseNet121	87.50	100.00	75.00	100.00	85.71	1.0000
MobileNetV2	87.50	100.00	75.00	100.00	85.71	1.0000
Equal-average fusion	87.50	100.00	75.00	100.00	85.71	1.0000
Manual weighted fusion	87.50	100.00	75.00	100.00	85.71	1.0000
Grid-optimized fusion	87.50	100.00	75.00	100.00	85.71	1.0000
Global trainable-weight fusion	87.50	100.00	75.00	100.00	85.71	1.0000
Sample-wise attention fusion	87.50	100.00	75.00	100.00	85.71	1.0000
Logistic-regression stacking	87.50	100.00	75.00	100.00	85.71	1.0000

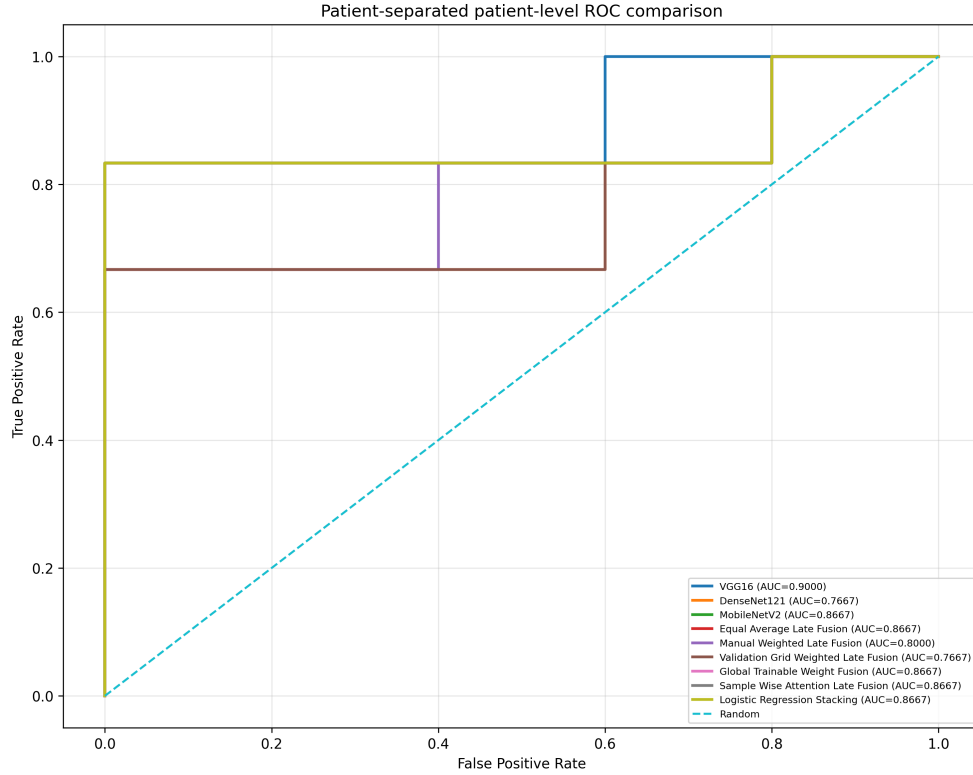


Fig. 5. Participant-level receiver operating characteristic curves for the six late-fusion strategies evaluated on the identical cohort of eight completely unseen participants. The curves overlap because all six fusion methods perfectly ranked the positive and negative participants.

Although the participant-level ROC-AUC reached 1.0000, the classification accuracy at the fixed decision threshold of 0.5 was 87.50%. This difference highlights the distinction between ranking performance and threshold-dependent classification performance. ROC-AUC evaluates the ability of the model to correctly rank positive and negative participants across all possible thresholds, whereas accuracy depends on a single operating threshold. Consequently, one participant with Parkinson's disease received a probability below the selected threshold, resulting in one false-negative prediction despite perfect ranking performance.

The participant-level evaluation represents a substantially more challenging validation protocol than the image-level experiment because all MRI slices belonging to a participant were assigned exclusively to the training, development, or testing subset. This protocol eliminates participant-level

overlap and provides a more reliable preliminary estimate of model generalization to previously unseen individuals. However, the participant-level test cohort contained only eight participants. Therefore, the reported participant-level results should be interpreted as preliminary evidence rather than definitive clinical validation. Future studies should evaluate the proposed framework using larger independent multi-center cohorts to assess its clinical applicability and generalization capability.

E. NTW Evaluation of the Selected Architectures

After the PPMI experiments were completed, selected best-performing architectures were initialized again using ImageNet-pretrained weights and independently trained using the NTW cohort. The following results therefore represent

separate NTW training and evaluation experiments rather than direct testing of the PPMI-trained model weights.

1) *NTW Image-Level VGG16 Performance:* Table XI presents the image-level performance of the frozen and fine-tuned VGG16 configurations on the held-out NTW test set containing 498 MRI slices.

Fine-tuning substantially improved VGG16 performance on the NTW cohort. Image-level accuracy increased from 76.51% with the frozen backbone to 87.95% after fine-tuning. Specificity increased from 52.31% to 72.22%, while recall increased from 95.04% to 100.00%. The ROC-AUC increased from 0.8739 to 0.9927, demonstrating that fine-tuning improved the ability of VGG16 to rank NTW Parkinson's disease and healthy-control slices.

Although the fine-tuned model correctly identified all Parkinson's disease slices, its specificity of 72.22% was lower than its recall. This difference indicates that some healthy-control slices were incorrectly classified as Parkinson's disease.

2) *NTW Participant-Level VGG16 Performance:* The participant-disjoint NTW test set contained 13 completely unseen participants, including six healthy controls and seven participants with Parkinson's disease. Table XII presents the participant-level VGG16 results.

Both VGG16 configurations achieved 61.54% participant-level accuracy, 100.00% recall, and 16.67% specificity. Each model correctly identified all seven participants with Parkinson's disease but correctly identified only one of the six healthy controls. The fine-tuned configuration improved ROC-AUC from 0.3333 to 0.7619 without changing the classifications produced at the fixed threshold of 0.5.

This result illustrates the difference between ROC-AUC and threshold-dependent classification performance. Fine-tuning improved the ordering of participant probabilities, but the fixed threshold continued to classify five healthy participants incorrectly. Consequently, the high sensitivity was obtained at the expense of low specificity.

3) *NTW Image-Level Early-Fusion Performance:* Two early-fusion backbone combinations were evaluated on the NTW cohort: VGG16 with DenseNet121 and MobileNetV2 with VGG16. Each combination was evaluated using feature concatenation and attention-based feature fusion.

Table XIII presents the NTW image-level performance of the four early-fusion configurations on the common test set of 498 MRI slices.

The VGG16-DenseNet121 concatenation model achieved the strongest overall image-level performance, with an accuracy of 93.98%, a precision of 90.65%, a recall of 99.65%, a specificity of 86.57%, an F1-score of 94.93%, and a ROC-AUC of 0.9942. Its confusion matrix contained 187 true negatives, 29 false positives, one false negative, and 281 true positives.

The VGG16-DenseNet121 attention model achieved 74.90% accuracy, an F1-score of 81.26%, and a ROC-AUC of 0.8924. Although its recall remained high at 96.10%, its specificity decreased to 47.22%, with 114 healthy-control slices incorrectly classified as Parkinson's disease.

For the MobileNetV2-VGG16 combination, concatenation achieved 89.56% accuracy, an F1-score of 91.48%, and a ROC-AUC of 0.9849. The attention configuration improved the accuracy to 91.77%, specificity to 83.33%, and F1-score to 93.11%, while obtaining a ROC-AUC of 0.9835.

Overall, VGG16-DenseNet121 concatenation produced the highest NTW image-level accuracy and ROC-AUC. The comparative ROC curves are shown in Fig. 8.

4) *NTW Participant-Level Early-Fusion Performance:* The early-fusion models were evaluated at the participant level on 13 completely unseen NTW participants, comprising six healthy controls and seven participants with Parkinson's disease.

The VGG16-DenseNet121 concatenation model achieved 53.85% accuracy, 53.85% precision, 100.00% recall, 0.00% specificity, an F1-score of 70.00%, and a ROC-AUC of 0.6190. It correctly classified all seven participants with Parkinson's disease but misclassified all six healthy controls.

The VGG16-DenseNet121 attention model achieved 61.54% accuracy, 58.33% precision, 100.00% recall, 16.67% specificity, an F1-score of 73.68%, and a ROC-AUC of 0.6905.

At a decision threshold of 0.4761, MobileNetV2-VGG16 concatenation achieved 61.54% accuracy, 62.50% precision, 71.43% recall, 50.00% specificity, an F1-score of 66.67%, and a ROC-AUC of 0.6667. At a threshold of 0.8628, MobileNetV2-VGG16 attention achieved 61.54% accuracy, 100.00% precision, 28.57% recall, 100.00% specificity, an F1-score of 44.44%, and a ROC-AUC of 0.4762.

Fine-tuned VGG16 achieved the highest participant-level ROC-AUC of 0.7619, whereas the highest participant-level accuracy among the early-fusion models was 61.54%. The comparative ROC curves are shown in Fig. 9. Because the test cohort contained only 13 participants, these results should be interpreted cautiously and validated using larger participant-disjoint cohorts.

As shown in Figure 10, the participant-level fusion model exhibited a widening difference between training and validation performance after fine-tuning began. Although the model fitted the training data strongly, this improvement did not translate into comparable validation or held-out participant performance. These findings reinforce the difference between slice-level classification and generalization to completely unseen participants.

F. Comparison with Existing Studies

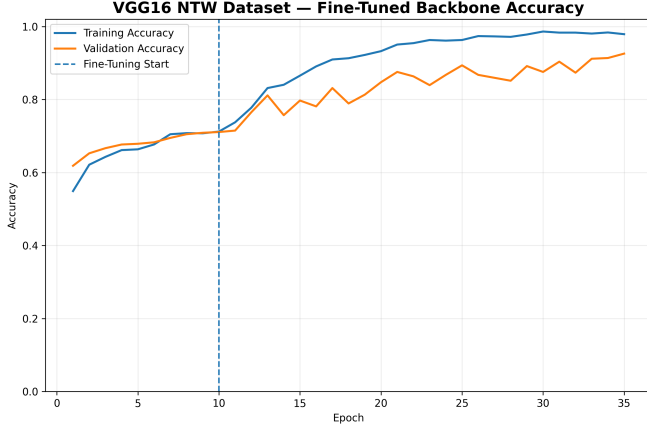
Published MRI-based Parkinson's disease studies have reported different performance levels under varying datasets, classification tasks, sample sizes, and evaluation protocols. Therefore, the reported results are presented for contextual comparison rather than direct ranking.

The proposed framework provides a comprehensive comparison of transfer learning, feature-level early fusion, and decision-level late-fusion approaches. Representative Grad-CAM visualizations from the fine-tuned VGG16 image-level model are included as a qualitative explainability analysis.

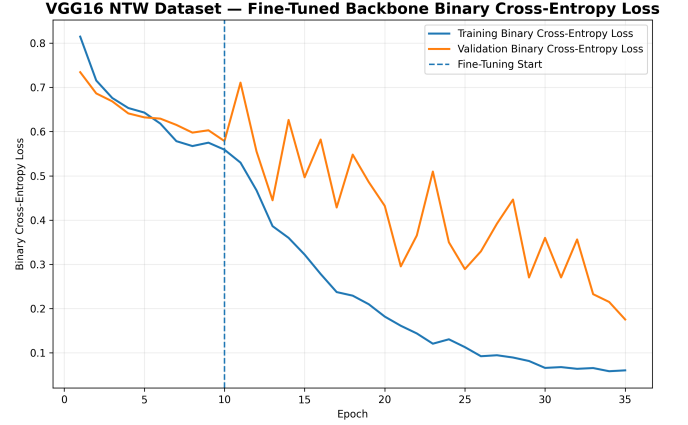
Table XIV compares the proposed framework with representative MRI-based Parkinson's disease classification studies.

TABLE XI
NTW IMAGE-LEVEL PERFORMANCE OF VGG16 ON 498 HELD-OUT MRI SLICES

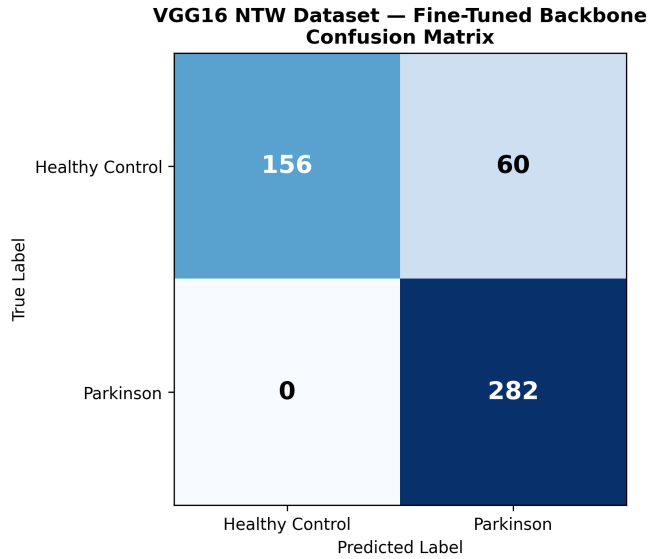
Configuration	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1-Score (%)	ROC-AUC
Frozen backbone	76.51	72.24	95.04	52.31	82.08	0.8739
Fine-tuned backbone	87.95	82.46	100.00	72.22	90.38	0.9927



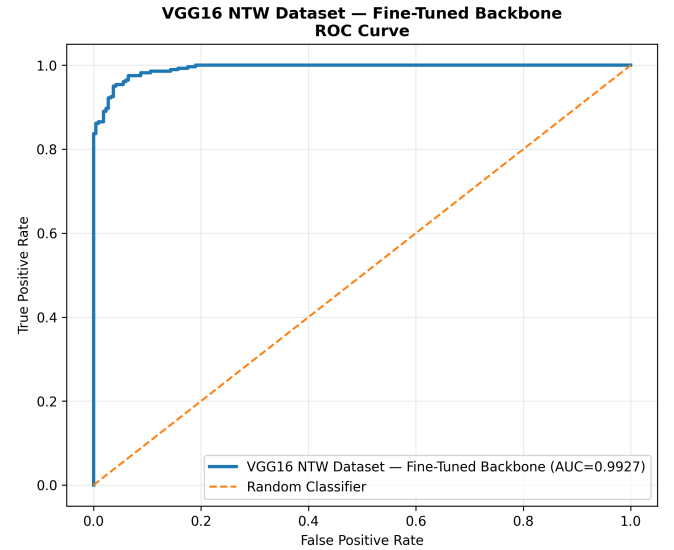
(a) Training and validation accuracy



(b) Training and validation loss



(c) Image-level confusion matrix



(d) Image-level ROC curve

Fig. 7. NTW image-level results for the fine-tuned VGG16 model. Panels (a) and (b) show the training and validation accuracy and binary cross-entropy loss across the frozen-backbone and fine-tuning stages. Panel (c) presents the confusion matrix for 498 held-out MRI slices, comprising 156 true negatives, 60 false positives, zero false negatives, and 282 true positives. Panel (d) shows the corresponding ROC curve with an AUC of 0.9927.

TABLE XII
NTW PARTICIPANT-LEVEL VGG16 PERFORMANCE ON 13 UNSEEN PARTICIPANTS

Configuration	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1-Score (%)	ROC-AUC	TN/FP/FN/TP
Frozen backbone	61.54	58.33	100.00	16.67	73.68	0.3333	1/5/0/7
Fine-tuned backbone	61.54	58.33	100.00	16.67	73.68	0.7619	1/5/0/7

The reported results are provided for contextual comparison rather than direct ranking because the studies used different datasets, participant populations, classification tasks, pre-processing procedures, and evaluation protocols.

The proposed framework achieved competitive image-level performance on PPMI, with VGG16 and grid-optimized late fusion reaching 98.97% accuracy. However, the participant-disjoint PPMI test cohort contained only eight participants.

TABLE XIII
NTW IMAGE-LEVEL PERFORMANCE OF THE EARLY-FUSION MODELS

Backbone Combination	Fusion Method	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1-Score (%)	ROC-AUC	TN/FP/FN/TP
VGG16 + DenseNet121	Concatenation	93.98	90.65	99.65	86.57	94.93	0.9942	187/29/1/281
VGG16 + DenseNet121	Attention	74.90	70.39	96.10	47.22	81.26	0.8924	102/114/11/271
MobileNetV2 + VGG16	Concatenation	89.56	85.06	98.94	77.31	91.48	0.9849	167/49/3/279
MobileNetV2 + VGG16	Attention	91.77	88.50	98.23	83.33	93.11	0.9835	180/36/5/277

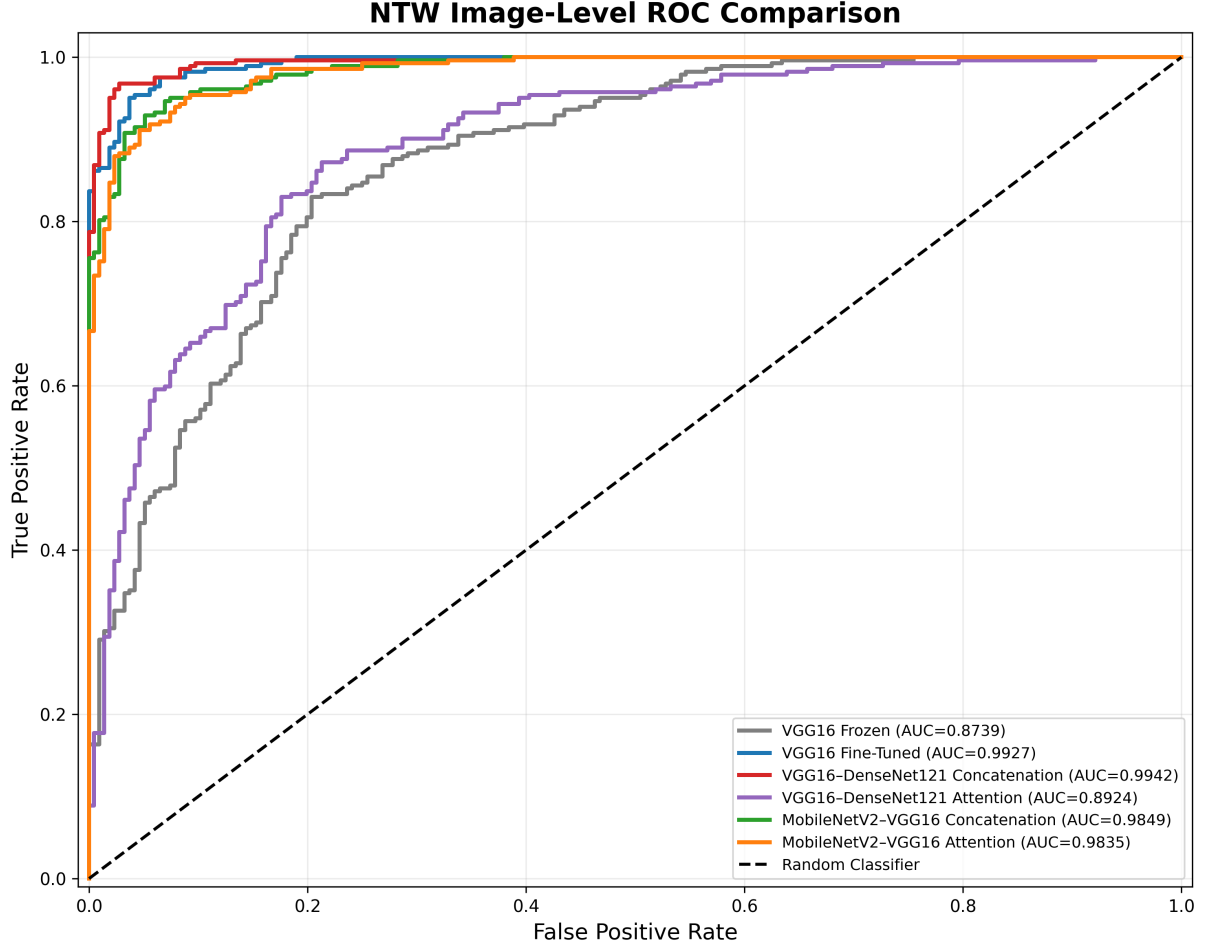


Fig. 8. ROC comparison of the NTW image-level models on 498 held-out MRI slices. The VGG16–DenseNet121 concatenation model achieved the highest ROC–AUC of 0.9942, followed by fine-tuned VGG16 with a ROC–AUC of 0.9927.

Therefore, the 87.50% participant-level accuracy should be interpreted as preliminary.

When the selected architectures were independently re-trained on the NTW cohort, the highest image-level accuracy was 93.98%, achieved by VGG16–DenseNet121 concatenation, with a ROC–AUC of 0.9942. The highest participant-level accuracy was substantially lower at 61.54%. This difference demonstrates why image-level results alone are insufficient for estimating generalization to completely unseen participants.

The main contribution of this study is therefore not only the reported classification accuracy but also the systematic separation of image-level and participant-level evaluation across two independent MRI cohorts. The framework additionally compares individual transfer-learning models, feature-level

early fusion, decision-level late fusion, and qualitative Grad-CAM explanations.

G. Limitations

Several limitations should be considered when interpreting the reported results. First, the image-level experiments divided individual MRI slices into training, validation, and testing subsets. Because multiple slices were generated from each three-dimensional MRI scan, slices from the same participant could occur in more than one image-level partition. Although the exact test slices were not used for training, participant-specific anatomical or acquisition characteristics may have contributed to the high image-level performance.

Second, the participant-disjoint test cohorts were small. The PPMI participant-level test set contained eight partici-

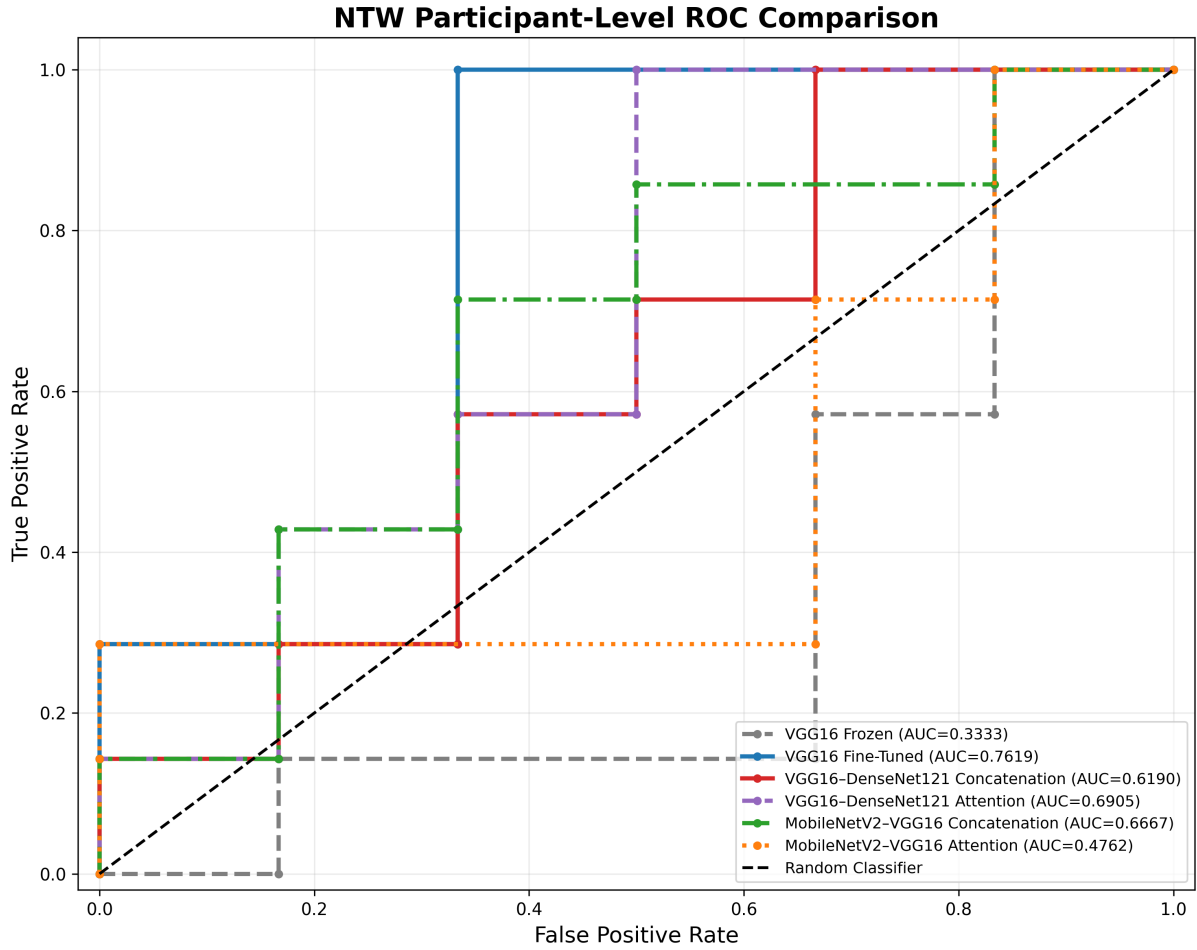


Fig. 9. ROC comparison of the NTW participant-level models on 13 completely unseen participants. Fine-tuned VGG16 achieved the highest ROC–AUC of 0.7619. These results should be interpreted cautiously because the test set contained only 13 participants.

TABLE XIV
COMPARISON OF THE PROPOSED METHOD WITH EXISTING MRI-BASED PARKINSON’S DISEASE CLASSIFICATION STUDIES

Study	Year	Dataset	Method	Reported Performance	Explainability
Camacho et al. [7]	2023	Multi-center T1 MRI	3D CNN using Jacobian maps	Accuracy: 79.3%; AUC: 0.87	Saliency maps
Dentamaro et al. [13]	2024	PPMI MRI and clinical data	DenseNet with EN intermediate fusion	F1-score: 96.5%	Integrated Gradients
Solana-Lavalle and Rosas-Romero [44]	2021	PPMI T1 MRI	Voxel-based morphometry with machine learning	Accuracy: 99.01% (men); 96.97% (women)	No
Basaia et al. [45]	2025	Multi-center T1 MRI	3D CNN using T1-weighted MRI	Accuracy: 74.04% ^b	CAM
Proposed Method	2026	PPMI and NTW T1 MRI	Transfer learning with early and late fusion	PPMI: 98.97% image level and 87.50% participant level; NTW: 93.98% image level and 61.54% participant level^a	Grad-CAM

^aFor PPMI, the 98.97% result corresponds to the matched image-level evaluation, where grid-optimized late fusion achieved a ROC–AUC of 0.9995. Under the PPMI participant-disjoint protocol, DenseNet121, MobileNetV2, and the evaluated late-fusion methods achieved 87.50% accuracy on eight unseen participants. For NTW, the selected architectures were independently initialized, retrained, and evaluated using NTW partitions. The VGG16–DenseNet121 concatenation model achieved 93.98% image-level accuracy and a ROC–AUC of 0.9942. At the participant level, the best NTW accuracy was 61.54% on 13 unseen participants. The NTW experiment was a separate second-cohort training experiment and did not directly test PPMI-trained weights on NTW images.

^bThe 74.04% result corresponds to classification of moderate-to-severe Parkinson’s disease participants versus controls using T1-weighted MRI alone. Results across studies are not directly comparable because the datasets, classification tasks, sample sizes, and evaluation protocols differ.

pants, while the NTW participant-level test set contained 13 participants. In the PPMI experiment, one additional error would change accuracy by 12.5 percentage points. In the NTW experiment, one additional error would change accu-

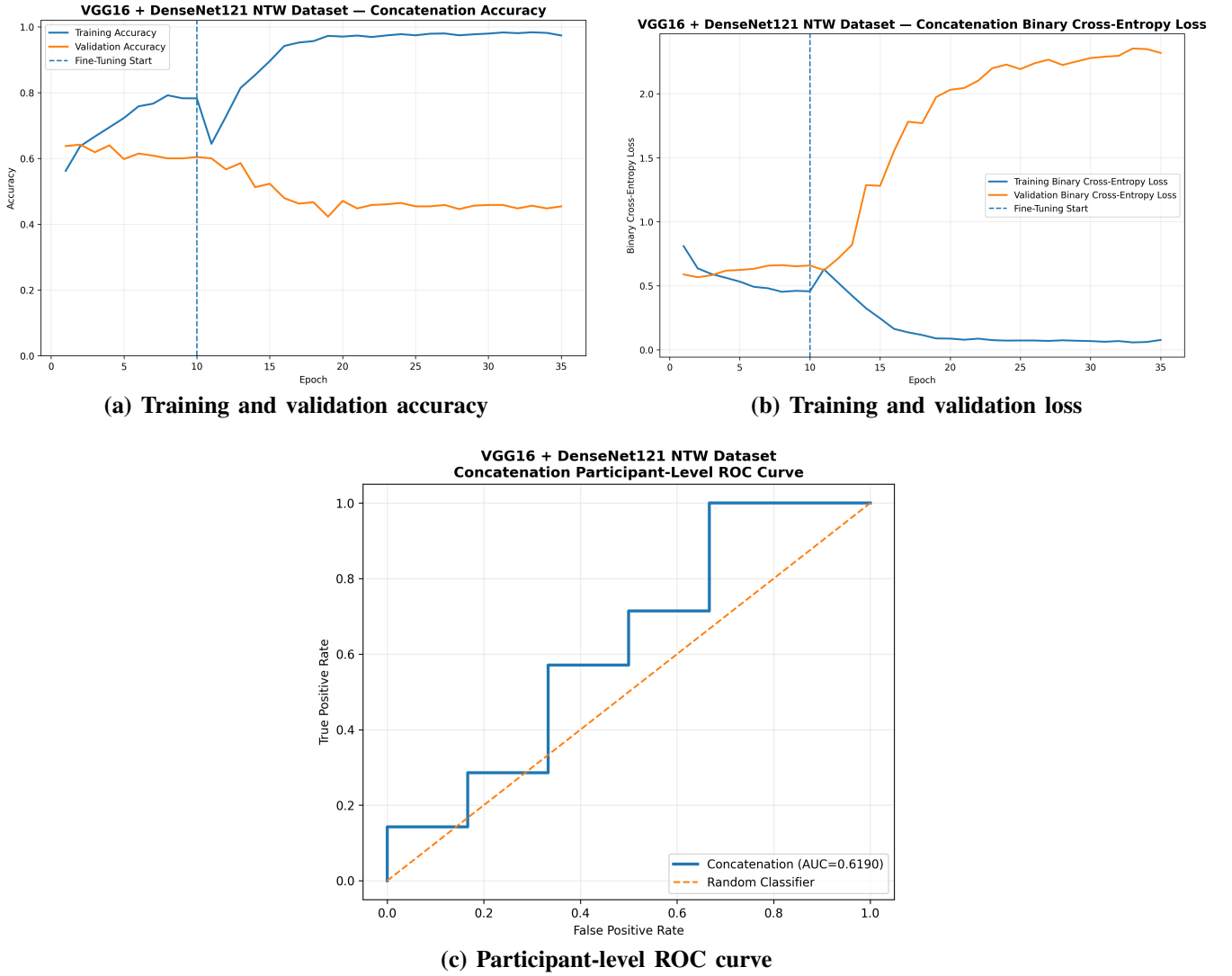


Fig. 10. Training behavior and participant-level discrimination of the VGG16–DenseNet121 concatenation model on the NTW cohort. Panels (a) and (b) show that training accuracy continued to increase and training loss decreased after fine-tuning, while validation accuracy declined and validation loss increased. This divergence indicates substantial overfitting under the participant-disjoint protocol. Panel (c) shows the ROC curve obtained from the 13 held-out participants, with an AUC of 0.6190.

racy by approximately 7.69 percentage points. Consequently, the participant-level metrics have substantial uncertainty and should not be interpreted as definitive estimates of clinical performance.

Third, the PPMI and NTW experiments were conducted as separate training experiments. The selected architectures were retrained using NTW training data; therefore, the NTW results do not represent direct external validation of fixed PPMI-trained models. A true external validation study would require freezing the PPMI-trained weights, preprocessing pipeline, calibration method, and classification threshold before evaluating an untouched external cohort.

Fourth, the conversion of three-dimensional MRI volumes into two-dimensional slices removed part of the volumetric spatial context. Individual slices from the same participant are also correlated and should not be treated as fully independent biological samples when interpreting statistical significance.

Fifth, the NTW cohort combined data from the NEURO-

CON and Tao Wu datasets, which were acquired using different scanners and imaging parameters. Although this heterogeneity provides a useful test of architecture reproducibility, scanner-related differences may influence the features learned by the models. Scanner harmonization and source-stratified analysis were not investigated in the present study.

Sixth, the NTW participant-level models exhibited substantial variation in threshold-dependent sensitivity and specificity. Fine-tuned VGG16 and VGG16–DenseNet121 attention achieved 100.00% recall but only 16.67% specificity, whereas VGG16–DenseNet121 concatenation produced 0.00% specificity. Conversely, MobileNetV2–VGG16 attention achieved 100.00% specificity but only 28.57% recall at its selected threshold. Probability calibration and threshold optimization may improve this balance, but thresholds must be selected using development data rather than the held-out test set.

Finally, the Grad-CAM visualizations provide qualitative explanations only. They do not demonstrate that the

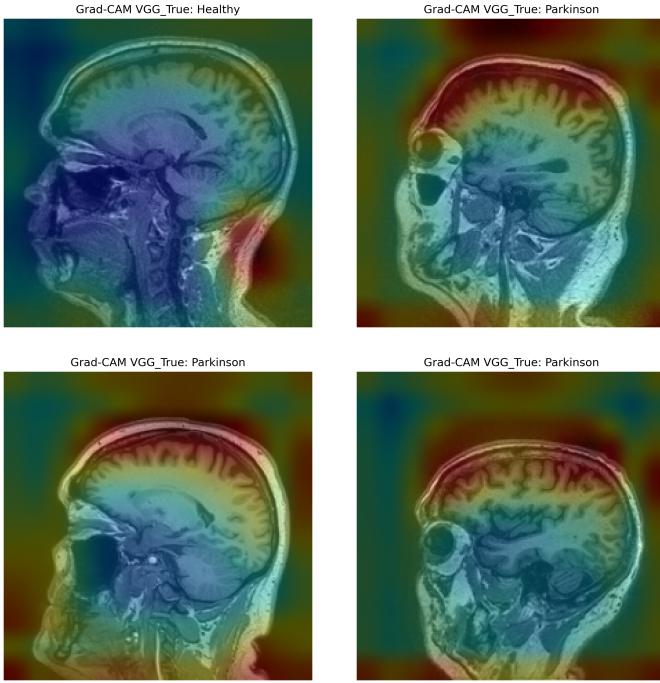


Fig. 11. Representative Grad-CAM visualizations generated from the fine-tuned VGG16 model. The first example corresponds to a healthy control, while the remaining three examples correspond to Parkinson's disease MRI slices. Warmer colors indicate image regions that contributed more strongly to the model's prediction.

highlighted regions represent validated Parkinson's disease biomarkers. Clinical and anatomical interpretation requires quantitative localization analysis and independent review by qualified neuroimaging specialists.

H. Grad-CAM Analysis

Grad-CAM was applied to the fine-tuned VGG16 image-level model to qualitatively visualize the regions influencing its classification predictions. Fig. 11 presents one representative healthy-control image and three representative Parkinson's disease images.

The highlighted regions indicate areas that contributed to the model's predictions. These visualizations are qualitative examples and should not be interpreted as clinical or anatomical evidence without further expert validation.

VI. CONCLUSION

This paper presented a two-cohort deep-learning study for Parkinson's disease classification using T1-weighted MRI. The primary PPMI experiments systematically compared VGG16, DenseNet121, and MobileNetV2 with feature-level early-fusion and decision-level late-fusion strategies. After the PPMI experiments, selected best-performing architectures were initialized again using ImageNet-pretrained weights, retrained using the NTW cohort, and evaluated using separate NTW test partitions.

Two evaluation protocols were investigated for each cohort. Image-level evaluation measured classification performance on held-out MRI slices, while participant-level evaluation

assigned all slices from each participant exclusively to one data partition. The participant-disjoint protocol therefore provided a stricter estimate of model performance on completely unseen participants.

In the matched PPMI image-level experiment, VGG16 achieved 98.97% accuracy, an F1-score of 98.96%, and a ROC-AUC of 0.9994. Grid-optimized late fusion matched VGG16's accuracy and achieved the highest PPMI image-level ROC-AUC of 0.9995. On eight completely unseen PPMI participants, DenseNet121, MobileNetV2, and the evaluated late-fusion methods achieved 87.50% accuracy and a ROC-AUC of 1.0000. However, the small participant test set limits the stability of this estimate.

In the independently trained NTW experiments, fine-tuned VGG16 achieved 87.95% image-level accuracy, an F1-score of 90.38%, and a ROC-AUC of 0.9927. The VGG16-DenseNet121 concatenation model achieved the highest NTW image-level accuracy of 93.98%, with an F1-score of 94.93% and a ROC-AUC of 0.9942.

The NTW participant-level results were substantially lower than the corresponding image-level results. The highest participant-level accuracy was 61.54% on 13 completely unseen participants. Fine-tuned VGG16 and VGG16-DenseNet121 attention achieved 100.00% recall but only 16.67% specificity, indicating a prediction bias toward the Parkinson's disease class. VGG16-DenseNet121 concatenation exhibited a stronger bias, classifying all six healthy controls incorrectly and producing 0.00% specificity. In contrast, MobileNetV2-VGG16 attention achieved 100.00% specificity but only 28.57% recall at its selected threshold, demonstrating that threshold-dependent sensitivity and specificity varied substantially among the models.

These findings demonstrate that strong image-level classification performance does not necessarily translate into reliable participant-level generalization. They also show that the relative performance of fusion strategies can vary across independent MRI cohorts. Attention improved the MobileNetV2-VGG16 image-level model on NTW but reduced the performance of VGG16-DenseNet121. Consequently, increased fusion complexity does not guarantee improved generalization.

Overall, transfer learning and fusion strategies demonstrated strong potential for MRI-based Parkinson's disease classification. Nevertheless, the participant-level results, small held-out cohorts, and variability between PPMI and NTW demonstrate that the developed models are not yet ready for clinical use. Larger multicenter participant-disjoint studies, probability calibration, and external testing of fixed trained models are required before the framework can be considered for clinical decision support. Grad-CAM provides useful qualitative insight into model predictions but should not be interpreted as anatomical or clinical evidence without expert validation.

VII. FUTURE WORK

Future work will evaluate the proposed framework using larger participant-disjoint test cohorts from multiple institutions. Confidence intervals and bootstrap analysis will be added to quantify the uncertainty associated with participant-

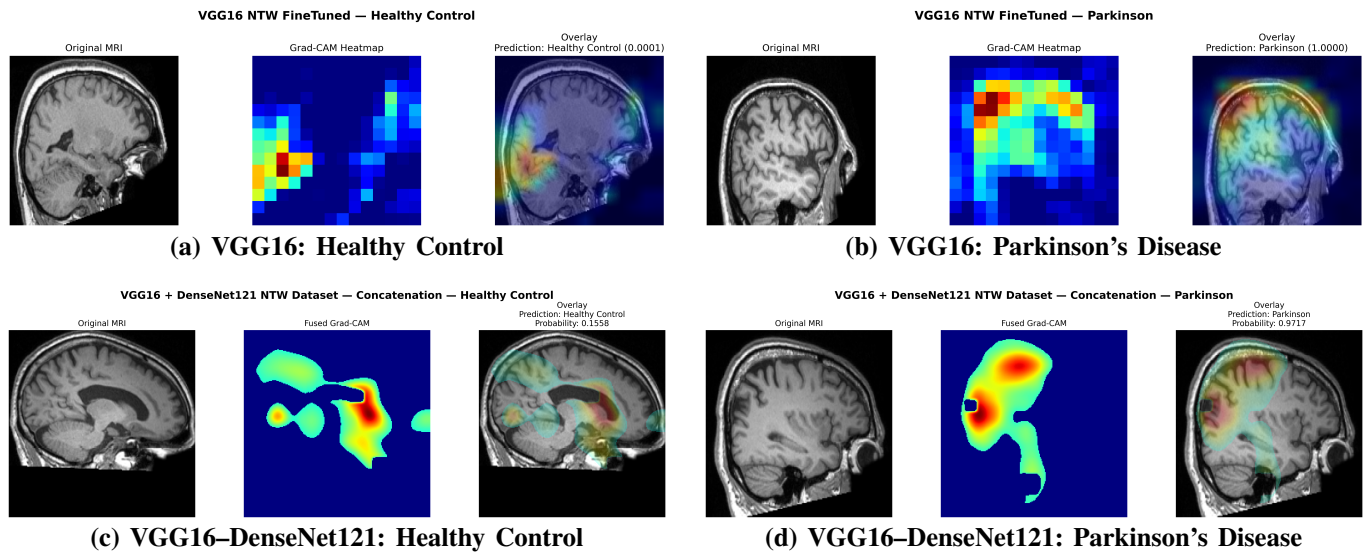


Fig. 12. Representative image-level Grad-CAM visualizations from the NTW cohort. Panels (a) and (b) show correctly classified healthy-control and Parkinson's disease slices from the fine-tuned VGG16 model. Panels (c) and (d) show qualitative fused activation maps generated from a separately retrained instance of the VGG16-DenseNet121 concatenation architecture. The separately retrained instance was used only for qualitative visualization, and its performance was not substituted for the original quantitative results reported in Table XIII. Warmer colors indicate regions contributing more strongly to each prediction. These maps should not be interpreted as anatomical segmentations or validated imaging biomarkers.

level performance, particularly when the number of test participants is small.

The current NTW experiment independently retrained the selected architectures using NTW data. A future direct external-validation experiment will freeze the complete PPMI-trained model, including its weights, preprocessing procedure, probability-calibration parameters, and classification threshold, and evaluate it on an untouched external dataset without retraining. This experiment will provide a more rigorous estimate of cross-dataset generalization.

Future research will also investigate scanner and intensity harmonization, domain adaptation, participant-level probability calibration, and threshold selection using development data. Three-dimensional CNNs and transformer-based architectures will be evaluated to preserve volumetric spatial information that may be lost when three-dimensional MRI scans are converted into independent two-dimensional slices.

The integration of multimodal clinical variables, including demographic information, motor assessments, cognitive measurements, and genetic biomarkers, may further improve participant-level classification. Advanced explainability methods will also be investigated together with quantitative localization analysis and expert clinical review. These developments are necessary to improve the reliability, interpretability, and clinical relevance of MRI-based Parkinson's disease classification systems.

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